

TechnoStore — Solution Blueprint

B2B Quote-to-Cash on Salesforce Revenue Cloud · DACH market reference architecture

1. Project context

TechnoStore GmbH is a B2B supplier of workstations, peripherals, cables, and software licenses to enterprise IT buyers across the DACH market. The reference implementation models the full Quote-to-Cash (Q2C) lifecycle on Salesforce Revenue Cloud (RLM + CLM) and Industries CPQ, integrating eight external systems for payment, logistics, e-signature, ticketing, messaging, enterprise data sync, and accounting.

2. Quality goals (priority order)

• Integration reliability

All external systems converge on a single Order activation within ~6 seconds, with HMAC-verified webhooks and idempotent retry handling.

• DACH market fit

19% German VAT visible at Quote stage, Sendcloud + DHL parcel routing, lexoffice invoicing, DATEV-konform CSV export, branded B2B PDFs.

• Architectural defensibility

Every significant decision documented as an Architecture Decision Record (ADR) with explicit alternatives considered.

• Maintainability

6-package SFDX layout, Kevin O'Hara TriggerHandler, Selector pattern, Mule-vs-Apex decision matrix.

• Reviewer scannability

README, ADRs, Mermaid diagrams, OpenAPI specs, Postman collection — each artifact answers a specific reviewer question in under 60 seconds.

3. Integrated external systems

Eight production-pattern integrations, all demo-verified end-to-end against vendor sandboxes:

- Stripe (payment intents + webhook on payment confirmation)
- DocuSign (bidirectional — outbound envelope from Apex + inbound Connect webhook -> Platform Event -> Contract status auto-update)
- Sendcloud / DHL (v3 Orders API for shipping label generation)
- JIRA (bidirectional ticketing; webhook also triggers Order activation)
- Slack (warehouse + ops channels)
- Notion (programmatically STAR-portfolio publishing)
- SAP S/4HANA (7-phase: ATP, sales-order ack, tax, CAMT.053 payment recon, master sync, CloudEvents webhook)
- Twilio WhatsApp (inbound message -> Salesforce Lead with regex email extraction)

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Architecture documentation, cross-cutting concepts, quality assurance

4. Cross-cutting concepts

- **Webhook security**

HMAC-SHA256 or shared-secret verification on every inbound endpoint; idempotency via dedicated event-log object.

- **Async after DML**

@future(callout=true) or Queueable for any callout after Approval.process() — prevents "uncommitted work pending" errors.

- **Privilege boundary crossing**

Site Guest Users publish Platform Events; triggers running under the Automated Process User perform record updates. The intended Salesforce pattern.

- **DACH finance**

Event-driven lexoffice invoicing on Stripe payment confirmation; DATEV-konform CSV export with SKR04 chart of accounts, German comma decimal, UTF-8 BOM.

- **Documentation as code**

31 Architecture Decision Records (Michael Nygard format), 5 Mermaid architecture diagrams, OpenAPI 3.0 specs, 51-entry STAR-format Notion portfolio.

5. Architecture artifacts

The system ships with a layered set of architecture artifacts. Each is scoped to a specific reviewer question:

- **arc42 Solution Blueprint**

Architect-readable overview (this document, full version 626 lines).

- **Architecture Decision Records**

31 ADRs covering: Mule-vs-Apex routing, Site + Guest User inbound pattern, idempotency, SAP integration phases, lexoffice + DATEV, and more.

- **Mermaid architecture diagrams**

C4 Context + Container + Q2C Sequence + Data Model + CI/CD pipeline.

- **OpenAPI 3.0 specifications**

Inbound webhook contracts published for every external system.

- **Notion STAR portfolio**

51 STAR-format case-study entries, generated programmatically via a custom Apex Notion-publishing service.

6. Quality assurance summary

Six-package SFDX layout. Kevin O'Hara TriggerHandler framework with recursion guards and bypass mechanism. Full Apex test coverage on every shipped class. FLS/CRUD enforcement via WITH USER_MODE on every read. Demo-verified end-to-end against vendor sandboxes — every flow has been observed working with real or sandbox traffic.